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Prognostic role of CD200 expression in hematologic malignancies: a systematic review and meta-analysis --Manuscript Draft--

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Abstract:	Summary Background: The predictability of CD200 in hematologic cancer patients remains controversial. This review aimed to study the prognostic effect of CD200 on the overall survival of patients with hematologic cancers. Methods: We searched PubMed, Scopus, Web of Science, Cochrane CENTRAL, Academic Search Complete, and CINAHL Complete for relevant publications on February 22, 2020. The eligible criteria for an included study were as follows: i) observational studies with the same management protocol across the study, ii) untreated human participants with any hematologic cancer, iii) having CD200 expression analysis, and iv) hazard ratios (HRs) or Kaplan-Meier curves for overall survival (OS) comparing CD200 positive (CD200+) with CD200 negative (CD200-) and high CD200 expression (hCD200) with low CD200 expression (lCD200) being available. Logarithmic hazards ratios (lnHRs) with corresponding standard errors (SEs) were pooled using a random-effect model to examine the prognostic role of CD200 on the survival outcomes. This study was registered at the Open Science Framework database (DOI: 10.17605/OSF.IO/HETSN). Findings: This review included 13 studies with 1,506 hematologic cancer patients. The meta-analysis showed that both CD200+ (lnHR = 0.07, 95% CI = 0.01–0.14) and hCD200 (lnHR = 0.47, 95% CI 0.22–0.71) negatively predicted the overall survival. The subgroup also found that both CD200+ (lnHR = 0.41, 95% CI 0.21–0.61) and hCD200 (lnHR = 0.72, 95% CI 0.34–1.11) negatively predicted to the overall survival rate of acute myeloid leukemia. Interpretation: CD200 is a negative prognostic factor of hematologic malignancies, especially acute myeloid leukemia. Funding: None

Summary

Background: The predictability of CD200 in hematologic cancer patients remains controversial. This review aimed to study the prognostic effect of CD200 on the overall survival of patients with hematologic cancers.

Methods: We searched PubMed, Scopus, Web of Science, Cochrane CENTRAL, Academic Search Complete, and CINAHL Complete for relevant publications on February 22, 2020. The eligible criteria for an included study were as follows: i(observational studies with the same management protocol across the study, ii(untreated human participants with any hematologic cancer, iii(having CD200 expression analysis, and iv(hazard ratios)HRs(or Kaplan-Meier curves for overall survival)OS(comparing CD200 positive)CD200+(with CD200 negative)CD200-(and high CD200 expression)hCD200(with low CD200 expression)lCD200(being available. Logarithmic hazards ratios)lnHRs(with corresponding standard errors)SEs(were pooled using a random-effect model to examine the prognostic role of CD200 on the survival outcomes. This study was registered at the Open Science Framework database (DOI: 10.17605/OSF.IO/HETSN).

Findings: This review included 13 studies with 1,506 hematologic cancer patients. The meta-analysis showed that both CD200+ (lnHR = 0.07, 95% CI = 0.01–0.14) and hCD200 (lnHR = 0.47, 95% CI 0.22–0.71) negatively predicted the overall survival. The subgroup also found that both CD200+ (lnHR = 0.41, 95% CI 0.21–0.61) and hCD200 (lnHR = 0.72, 95% CI 0.34–1.11) negatively predicted to the overall survival rate of acute myeloid leukemia.

Interpretation: CD200 is a negative prognostic factor of hematologic malignancies, especially acute myeloid leukemia.

Funding: None

Research in context

Evidence before this study

CD200, otherwise known as OX-2 membrane glycoprotein, is an immune checkpoint which suppresses immune function. Not only is CD200 expressed on a wide variety of tissues all over the body, but the protein can also be found on cancer cells. A plethora of studies reported CD200 as a predictor of poor survival rate in cancer. However, there has never been a systematic synthesis to validate such phenomenon. We did a systematic review to examine the evidence from observational studies regarding the prognostic role of CD200 in hematologic malignancies. We searched PubMed, Scopus, Web of Science, Cochrane CENTRAL, Academic Search Complete, and CINAHL Complete for relevant publications from their inception to February 22, 2020. There was no limitation on clinical setting, date, language, or literature type. We identified 13 studies, comprising of acute myeloid leukemia, multiple myeloma, pediatric B-cell acute lymphoblastic leukemia, adult B-cell acute lymphoblastic leukemia, myelodysplastic syndrome, and chronic lymphocytic leukemia. Meta-analyses of 1506 participants were conducted alongside with the assessment of bias, following PRISMA guidelines.

Added value of this study

To our knowledge, this study is the first published meta-analysis regarding the prognostic significant CD200 and hematologic malignancies. The pooled outcome of our study demonstrated that CD200 positivity and high level of CD200 expression are negative prognostic factors of overall survival in hematologic cancer patients, especially in acute myeloid leukemia.

Implications of all the available evidence

CD200 has been studied more rigorously in these recent years. There are pre-clinical studies, observational studies, and even clinical trials exploring the possibilities of CD200 as a target for cancer therapy. Meta-analyses of these studies suggested that CD200 serves as a negative prognostic factor in hematologic malignancies. Further studies are warranted to find mechanisms underlying such phenomenon and examine the possibility of using CD200 as a target for hematologic malignancies treatments.

Introduction

Hematologic malignancies are cancers that originated from hematopoietic cells. The American Cancer Society has estimated that there would be 176,200 new cases and 56,770 deaths from hematologic cancers.¹ Despite recent advances in hematologic malignancies, improving the survival rate of such diseases is still a challenging task. Not only could the identification of new prognostic factors help clinicians make their decisions on treatment strategies, but it could also pave ways to new targeted cancer therapies.

CD200 or OX-2 membrane glycoprotein is expressed in many organs in the body. It has been identified as an immune suppressor in various types of cancer, such as glioma and bladder cancer.^{2,3} The suppression mechanisms include suppressing the secretion of pro-inflammatory cytokines and enhancing regulatory T-cell activity.⁴⁻⁶ Despite a plethora of studies exploring the prognostic significance of CD200 in various cancer types, there has never been an attempt to explore the prognostic impact of CD200 on hematologic malignancies. This review aimed to study the prognostic effect of CD200 on the overall survival of patients with hematologic malignancies.

Methods

Search Strategy and selection criteria

We performed systematic searches to retrieve relevant publications from electronic databases, including Pubmed, Scopus, Web of Science, Cochrane Library, Academic Search Complete, and CINAHL Complete. The search terms for the database searches were provided in the supplementary data (*Appendix 1*). We identified all the relevant publications until February 22, 2020. The identified publications were imported to the Zotero program with the duplicated citations excluded. The non-duplicated publications were then imported to the Rayyan QCRI website to be assessed for their eligibility.⁷

The eligible criteria for the included studies were as follows: i) observational studies with the same management protocol across the study, ii) untreated human participants with any hematologic cancer, iii) having CD200 expression analysis, and iv) hazard ratios (HRs) or Kapan-Meier curves for overall survival (OS) comparing CD200 positive (CD200+) with CD200 negative (CD200-) and high CD200 expression (hCD200) with low CD200 expression (lCD200) being available.

Two reviewers (K. S. and T. T.) independently screened the abstracts and evaluated the full texts of the publications imported to the Rayyan QCRI website. Each reviewer's decisions were blinded from another reviewer until both reviewers completed the inclusion of eligible studies. Any discrepancy was resolved by a consensus discussion with a third-party reviewer (C.K). For discrepancies that required further information that was not reported in the full text, we contracted the corresponding author of the publication for clarification. Neither language restriction nor publication date limitation was applied. The data of the included studies were then extracted as follows: i) study ID (first author, publication year), ii) characteristics of the participants (age, gender, cancer type, ethnic/country, and follow-up duration), iii) CD200 analysis (analysis method and cut-off points of CD200 expression), and iv) hazard ratios (HRs) and corresponding standard errors (SEs) of OS comparing CD200+ with CD200- or hCD200 with lCD200. For included studies that did not report the HRs, we extracted the data from the Kaplan-Meier curves and calculated their HRs from the extracted data using the methods from previous publications.⁸

The Quality in Prognosis Studies (QUIPS) tool was used to assess the quality of the included studies.⁹ Low risk of bias studies was defined in all domains. Moderate risk of bias studies was studied with one or more domains with a moderate risk of bias but none of the domains was rated as high risk of bias. Studies having at least one domain with a high risk of bias were classified as high risk of bias studies. Two reviewers (K.S. and C.K.) independently evaluated

the quality of the included studies. Any discrepancy was resolved by a consensus discussion with a third-party reviewer (T.T.).

Data analysis

We performed the meta-analyses of pooled logarithmic hazards ratios (lnHRs) to estimate the prognostic impact of CD200 on hematologic malignancies in terms of OS. An HR higher than 0 was considered as a negative prognosis. The lnHRs and SEs were pooled using the inverse variance method. Meta-analyses with I^2 lower than 50% indicated the use of the fixed-effect model. On the contrary, the random effect model was applied to the meta-analyses with their I^2 equals to or higher than 50%. The DerSimonian-Laird method was used to estimate the τ^2 . The reported 95% confidence intervals (CIs) of the HRs were transformed into SEs using the following formula:

$$SE = \frac{(\ln(\text{upper } 95\% \text{ CI}) - \ln(\text{lower } 95\% \text{ CI}))}{2 \phi^{-1}(1 - \alpha/2)}$$

where ϕ^{-1} is the inverse function of the cumulative distribution of the normal distribution and α is the significance level of type I error, which was set at 0.05 in this study.

The Egger's test for funnel plot asymmetry was performed to explore the publication bias. The p-value was lower than 0.05 inferred statistical significance.

All the analyses were carried out by using the Rstudio software version 1.2.5033.¹⁰ For studies without HRs, the data from the Kaplan-Meier curves were extracted by Engauge Digitizer software version 6.0.¹¹ The protocol of this study was prospectively registered at the Open Science Framework database (DOI: 10.17605/OSF.IO/HETSN).

Role of the funding source

There was no funding source for this study. The funding sources of all study authors played no role in study design, data collection, data analysis, data interpretation, or writing of the report. The corresponding author had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Results

Literature Search

We identified 221 publications from the database searches. One hundred thirteen non-duplicated publications were included for the eligibility evaluation. After the abstract screening and full-text assessment, a total number of 1,506 participants from 13 studies were included in the meta-analyses (Figure 1).^{5, 12-23}

Figure 1: The PRISMA flow diagram of the literature search.

Study characteristics and quality assessment

Thirteen studies were included consisting of five acute myeloid leukemia (AML) studies, four multiple myeloma (MM) studies, one pediatric B-cell acute lymphoblastic leukemia (PBALL) study, one adult B-cell acute lymphoblastic leukemia (ABALL) study, 1 myelodysplastic syndrome (MDS) study, and one chronic lymphocytic leukemia (CLL) study (*Table 1*). Of 13 studies, five studies were conducted in Europe, four studies were carried out in Africa, and 4 studies were performed in Asia. The publication years of the studies ranged from 2007 to 2019. By using the QUIPS tool (*Table 2*), six studies had a low risk of bias and the other seven studies had a moderate risk of bias. The moderate risk of bias resulted from the lack of multivariate analysis.

Prognostic role of CD200 positivity in hematologic malignancies (Figure 2)

We explored the prognostic role of CD200 positivity in hematologic malignancies. There were 1365 participants from 11 studies included in our meta-analysis. The pooled lnHR showed that CD200 positivity is a predictor of reduced overall survival in hematologic malignancies (lnHR = 0.07, 95% CI = 0.01–0.14). However, the pooled outcome was heterogeneous ($I^2 = 70\%$), so the random-effect model was applied to our meta-analysis. Subgroup analysis based on cancer type demonstrated that CD200 positivity maintained its negative prognostic role in AML (lnHR = 0.41, 95% CI 0.21–0.61, 4 studies), ABALL (lnHR = 0.02, 95% CI 0.00–0.04, 1 study), and PBALL (lnHR = 0.05, 95% CI 0.03–0.07, 1 study). On the contrary, CD200 positivity was not associated with higher mortality in MM and CLL. No study explored the prognostic impact of CD200 positivity on the overall survival of hematologic cancer patients. Significant publication bias was not found using the Egger's test for funnel plot asymmetry ($p = 0.06$) (*Appendix 2, supplementary data*).

Figure 2: The forest plot of the meta-analysis of the logarithmic hazard ratio comparing CD200 positive and CD200 negative hematologic cancer patients.

Remark: Zahran et al. 2018 did not explore the prognostic impact of CD200 positivity on the overall survival of hematologic cancer patients.

ABALL: adult B-cell acute lymphoblastic leukemia; AML: acute myeloid leukemia; CLL: chronic lymphoblastic leukemia; MM: multiple myeloma; PBALL: pediatric B-cell acute lymphoblastic leukemia

Prognostic role of CD200 level of expression in hematologic malignancies (Figure 3)

The prognostic role of the CD200 level of expression in hematologic malignancies was explored. Eight hundred thirty-one participants from five studies were included in our meta-analysis. The pooled lnHR showed that hCD200 is a negative prognostic factor in hematologic malignancies (lnHR = 0.47, 95% CI 0.22–0.71). The fixed-effect model was applied to our meta-analysis as the pooled outcome was not heterogeneous ($I^2 = 41\%$). Subgroup analysis based on cancer type demonstrated that hCD200 expression maintained its negative prognostic role of overall survival in AML (lnHR = 0.72, 95% CI 0.34–1.11, 3 studies) and MDS (lnHR = 1.19, 95% CI 0.15–2.23, 1 study), but not in CLL. The publication bias was not explored due to the inadequate number of studies.

Figure 3: The forest plot of the meta-analysis of the logarithmic hazard ratio comparing hematologic cancer patients with high CD200 expression and low CD200 expression. AML: acute myeloid leukemia; CLL: chronic lymphoblastic leukemia; MDS: myelodysplastic syndrome

Discussion

CD200 immune checkpoint plays an important role in the prognostication of various cancers, including hematologic cancers. Several studies demonstrated the negative prognostic role of CD200 on acute myeloid leukemia clinical outcomes, including both overall survival and disease-free survival.^{5,12,14,17} There were a few proposed explanations of this phenomenon. First, CD200 expressed on tumor cells could suppress activities of immune cells, such as memory T-cell and natural killer cells.^{4,24} Second, CD200 could decrease pro-inflammatory cytokines secretion mediated through AML-dendritic cells (DC) mixed lymphocyte reactions (MLRs), such as IL-12 and IFN- γ , and increase anti-inflammatory cytokine secretion through monocyte-DC MLRs, such as TGF- β and IL-10.²⁵ Third, CD200 could enhance the function of regulatory T-cells, especially FoxP3 positive regulatory T-cells.^{6,25} Also, CD200 expression is positively related to programmed death-ligand 1 (PD-L1) expression, a well-known immune checkpoint.²⁶ One study also showed that high expression level of CD200 was a risk factor of disease progression from MDS to AML.¹⁸ From current data, it is possible to conclude that CD200 is a negative prognostic factor in AML. Despite several lines of evidence regarding the negative impact of CD200 on clinical outcomes of cancer patients, some studies reported controversial results. In ALL, CD200 expression was associated with reduced overall survival.^{19,20} However, one study demonstrated that CD200 overexpression is associated with excellent clinical outcomes in pediatric B-cell ALL.²⁷ Moreover, the prognostic impact of CD200 in MM remained controversial. A few studies demonstrated reduced overall survival^{13,15,21}, but Moreaux et al. reported that CD200 expression is positively associated with better event-free survival.²⁸ Clinical trials exploring CD200 blockade are still limited. A phase I clinical trial studying the effect of Samalizumab, a CD200 monoclonal antibody, on CLL and MM reported that blocking CD200 had a positive effect on CLL, but not MM.²⁹

This is the first meta-analysis to demonstrate the negative prognostic role of CD200 in hematologic malignancies. However, we encountered several limitations and thus, interpretation of the results must be done with caution. First, the number of tumor types in this study was limited. There was no lymphoma study and therefore, the generalizability of the results might be reduced. Second, the CD200 analysis method and the cut-off value defining CD200 positivity and high CD200 expression varied among studies. Third, we estimated the hazard ratio from a few studies, which could affect the accuracy of the result. Finally, we encountered a small number of studies in a few cancer types.

Our results suggest that CD200 is a negative prognosis factor in hematologic malignancies. The evidence of the negative prognostic impact seems to be strong in AML. However, there are still gaps in knowledge regarding CD200 and hematologic malignancies. More studies are warranted to validate the effect of CD200 in hematologic malignancies.

Contributors

K.S., T.T., and C.K.: Conceptualization, data curation, methodology, investigation, formal analysis, writing-original draft, and writing-review and editing.

T.R.: Conceptualization, supervision, validation, and writing-review and editing.

All authors critically reviewed the manuscript and provided important intellectual contributions. The final manuscript was approved by all authors.

Declaration of Interests

We declare no competing interests.

Acknowledgments

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Figure1

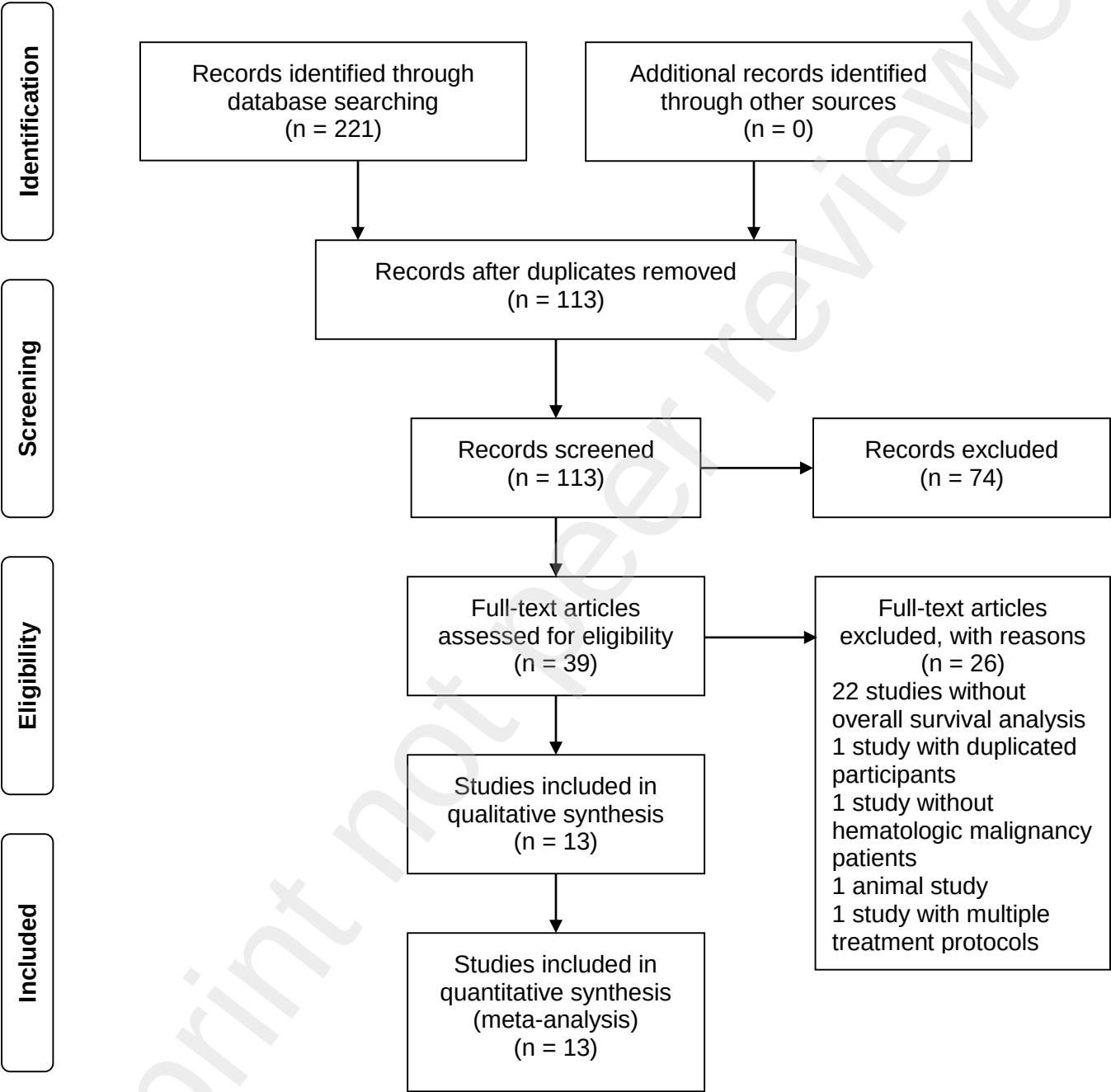


Figure2

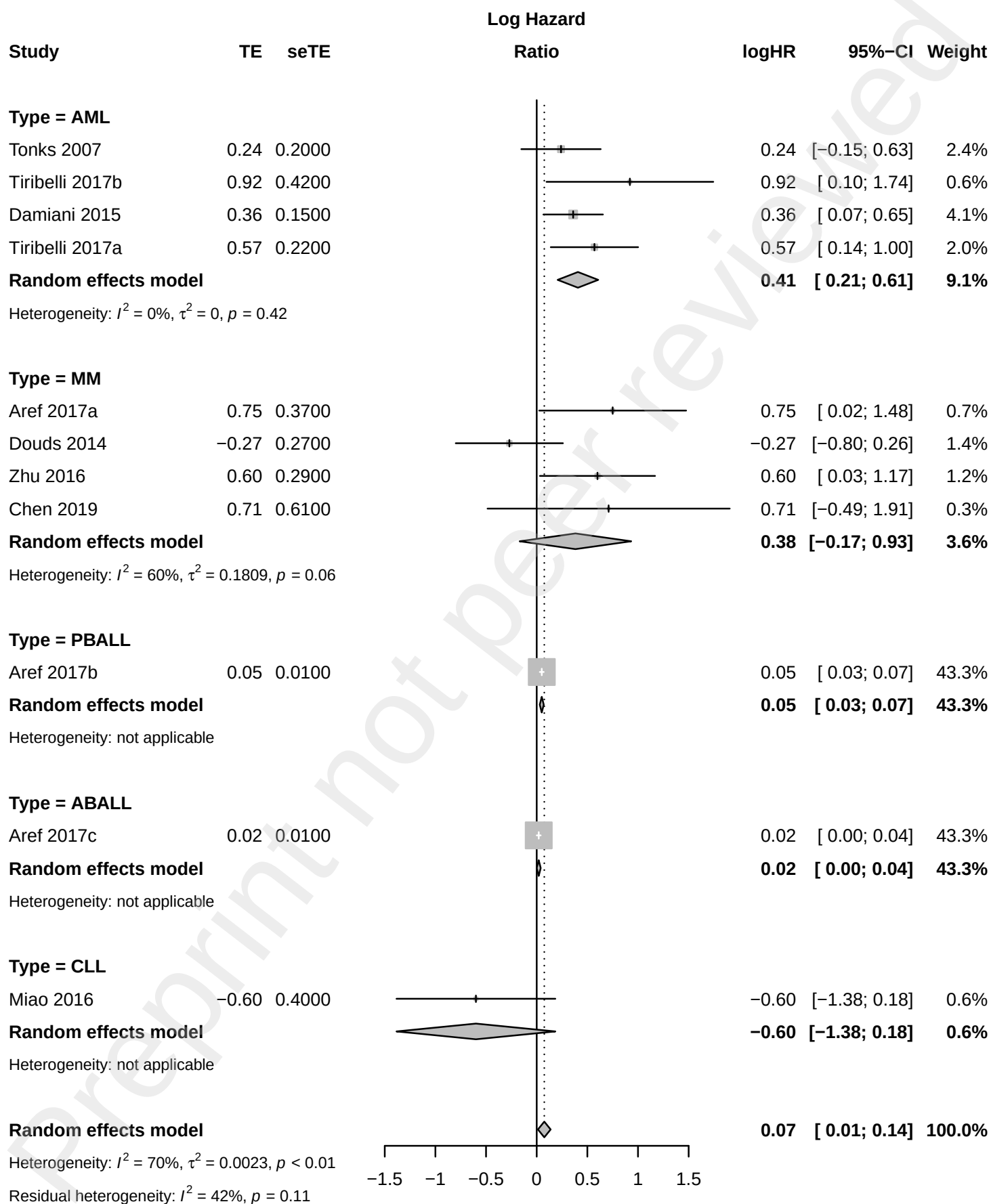


Figure3

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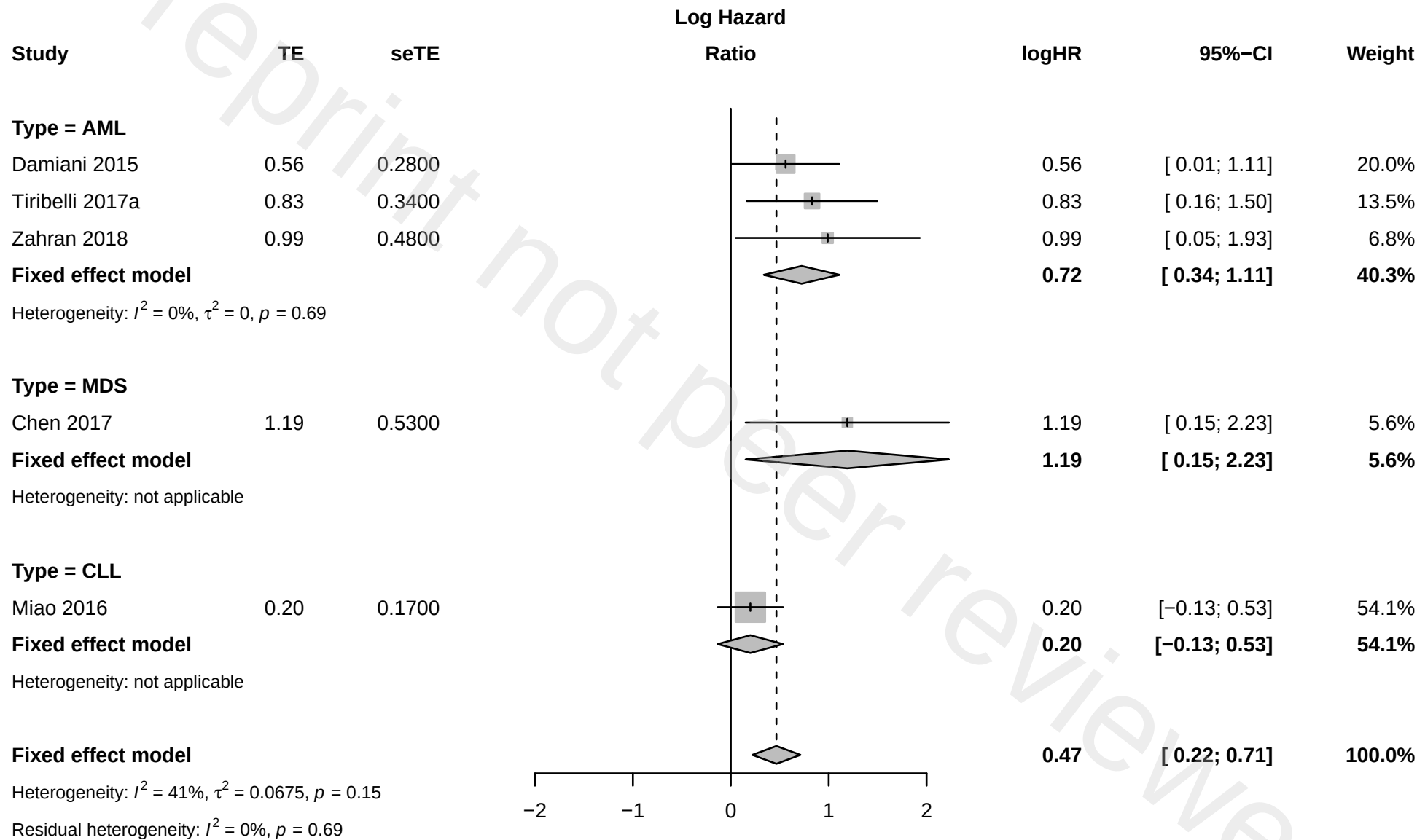


Table 1: Study Characteristics

Study	Characteristics of the participants	CD200 analysis	Hazard ratio
Aref 2017a	Egypt 50 MM patients (M/F = 0.36) Median age: 57.2 years old Follow-up duration: NA	-Flow cytometry -CD200 expression of plasma cells -CD200+ if $\geq 20\%$ of plasma cells express CD200 exceeding the isotype control threshold	CD200+ vs. CD200- (estimated)
Aref 2017b	Egypt 43 PBALL patients (M/F = 0.58) Median age: 5.4 years old Follow-up duration: 24 months	-Flow cytometry -CD200 expression of leukemic blasts -CD200+ if there is more than 1 log shift in MFI compared to isotype control	CD200+ vs. CD200- (reported, multivariate)
Aref 2017c	Egypt 70 ABALL patients (M/F = 0.57) Median age: NA Follow-up duration: NA	-Flow cytometry -CD200 expression of bone marrow blast cells -CD200+ if $\geq 20\%$ of bone marrow blast cells express CD200 marker	CD200+ vs. CD200- (reported, multivariate)
Chen 2017	China 101 MDS patients (M/F = 0.50) Median age: 65.4 years old Follow-up duration: 45 months	-Flow cytometry -CD200 expression of CD33+ bone marrow cells -hCD200 if CD200 expression is higher than the median expression level	hCD200 vs. lCD200 (reported, multivariate)
Chen 2019	China 131 MM patients (M/F = NA) Median age: NA Follow-up duration: 30 months Grey Literature	-Flow cytometry -CD200 expression of bone marrow cells -CD200+ if $\geq 20\%$ of bone marrow cells express CD200 marker	CD200+ vs. CD200- (estimated)
Damiani 2015	Italy 244 AML patients (M/F = 0.71) Median age: 59 years old Follow-up duration: NA	-Flow cytometry -CD200 expression of blast cells -CD200+ if $\geq 20\%$ of blast cells express CD200 exceeding the isotype control threshold -hCD200 if MFI > 11	CD200+ vs. CD200- (estimated) hCD200 vs. lCD200 (reported, multivariate)

ABALL: adult B-cell acute lymphoblastic leukemia; AML: acute myeloid leukemia; CD200+: CD200 positive; CD200-: CD200 negative; CLL: chronic lymphoblastic leukemia; hCD200: high CD200 expression; lCD200: low CD200 expression; MDS: myelodysplastic syndrome; MFI: mean fluorescence intensity; MM: multiple myeloma; M/F: male to female ratio; PBALL: pediatric B-cell acute lymphoblastic leukemia; NA: Not available

Table 1: Study Characteristics (continued)

Study	Characteristics of the participants	CD200 analysis	Hazard ratio
Douds 2014	The United States of America 54 MM patients (M/F = 0.65) Median age: 62 years old Follow-up duration: NA	-Immunohistochemistry staining -CD200 expression of bone marrow biopsies -CD200+ if scored as either positive or negative by using moderate to strong membranous staining in 25% of plasma cells as a cut-off by three pathologists	CD200+ vs. CD200- (estimated)
Miao 2016	China 307 CLL patients (M/F = 0.66) Median age: 61 years old Follow-up duration: 51 months	-Flow cytometry -CD200 expression of CD19+ cells -CD200+ if $\geq 30\%$ of CD19+ cells express CD200 marker -hCD200 if MFI > 189.5	CD200+ vs. CD200- (estimated) hCD200 vs. lCD200 (estimated)
Tiribelli 2017a	Italy 139 AML patients (M/F = 0.58) (cytogenetically normal) Median age: 60 years old Follow-up duration: 18 months	-Flow cytometry -CD200 expression of blast cells -CD200+ if $\geq 20\%$ of blast cells express CD200 exceeding the isotype control threshold -hCD200 if MFI > 11	CD200+ vs. CD200- (estimated) hCD200 vs. lCD200 (reported, multivariate)
Tiribelli 2017b	Italy 65 AML patients (M/F = 0.71) (ELN favorable subgroup) Median age: 57 years old Follow-up duration: NA	-Flow cytometry -CD200 expression of blast cells -CD200+ if $\geq 20\%$ of blast cells express CD200 exceeding the isotype control threshold -hCD200 if MFI > 11	CD200+ vs. CD200- (reported, multivariate)
Tonks 2007	The United Kingdom 184 AML patients (M/F = NA) Median age: 49.6 years old Follow-up duration: NA	-Oligonucleotide arrays -CD200 gene expression -CD200+ if CD200 gene expression is positive for oligonucleotide array analysis	CD200+ vs. CD200- (reported, univariate)

ABALL: adult B-cell acute lymphoblastic leukemia; AML: acute myeloid leukemia; CD200+: CD200 positive; CD200-: CD200 negative; CLL: chronic lymphoblastic leukemia; hCD200: high CD200 expression; lCD200: low CD200 expression; MDS: myelodysplastic syndrome; MFI: mean fluorescence intensity; MM: multiple myeloma; M/F: male to female ratio; PBALL: pediatric B-cell acute lymphoblastic leukemia; NA: Not available

Table 1: Study Characteristics (continued)

Study	Characteristics of the participants	CD200 analysis	Hazard ratio
Zahran 2018	Egypt 40 AML patients (M/F = NA) (cytogenetically normal) Median age: 35 years old Follow-up duration: NA	-Flow cytometry -CD200 expression of myeloid blasts -hCD200 if $\geq 50\%$ of myeloid blasts express CD200 marker	hCD200 vs. lCD200 (estimated)
Zhu 2016	China 78 MM patients (M/F = 0.54) Median age: 63 years old Follow-up duration: 30 months	-Flow cytometry -CD200 expression of plasma cells -CD200+ if $\geq 20\%$ of plasma cells express CD200 marker	CD200+ vs. CD200- (estimated)

ABALL: adult B-cell acute lymphoblastic leukemia; AML: acute myeloid leukemia; CD200+: CD200 positive; CD200-: CD200 negative; CLL: chronic lymphoblastic leukemia; hCD200: high CD200 expression; lCD200: low CD200 expression; MDS: myelodysplastic syndrome; MFI: mean fluorescence intensity; MM: multiple myeloma; M/ F: male to female ratio; PBALL: pediatric B- cell acute lymphoblastic leukemia; NA: Not available

Table 2: Study Quality

Study	Study Participation	Study Attrition	Prognostic Factor Measurement	Outcome Measurement	Study Confounding	Statistical Analysis and Reporting
Aref 2017a	LR	LR	LR	LR	MR	LR
Aref 2017b	LR	LR	LR	LR	LR	LR
Aref 2017c	LR	LR	LR	LR	LR	LR
Chen 2017	LR	LR	LR	LR	LR	LR
Chen 2019	LR	LR	LR	LR	MR	LR
Damiani 2015	LR	LR	LR	LR	LR	LR
Douds 2014	LR	LR	LR	LR	MR	LR
Miao 2016	LR	LR	LR	LR	MR	LR
Tiribelli 2017a	LR	LR	LR	LR	LR	LR
Tiribelli 2017b	LR	LR	LR	LR	LR	LR
Tonks 2007	LR	LR	LR	LR	MR	LR
Zahrn 2018	LR	LR	LR	LR	MR	LR
Zhu 2016	LR	LR	LR	LR	MR	LR

LR: low risk of bias; MR: moderate risk of bias

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